Capstone 6 – Error Log

Records wrong explanations, outdated syntax and unsafe code suggestions found while checking AI explanations against real runs in Google Colab.

How to use: Rows E1–E5 were confirmed by running the code (document 2). Rows E6–E10 are the typical outdated/unsafe patterns that the scanner catches. In the **Tool that said it** column write the AI tool (ChatGPT, Gemini, Claude) and date **only if you saw that tool produce the mistake**; otherwise leave it as "not observed". Never claim a tool made an error it did not make.

ID	Program	Category	Wrong or risky statement	What really happens	Correct explanation / fix	Severity	Tool that said it
E1	P01	Wrong explanation	`range(1, 6)` includes 6	Stops at 5; end value is excluded	Say "1 up to but not including 6"	Medium	*fill in*
E2	P03	Wrong output claim	factorial(0) raises an error or returns 0	Returns 1 because the loop never runs	0! = 1 by definition and by the code	Medium	*fill in*
E3	P04	Wrong output claim	`fruits[1:3]` is `['banana', 'cherry']`	`['blueberry', 'banana']` (insert shifted items)	Trace the list after every edit before slicing	High	*fill in*
E4	P08	Wrong explanation	Mode `"a"` overwrites the file	Appends to the end	`"w"` overwrites, `"a"` appends	High	*fill in*
E5	P09	Wrong output claim	Each call gets a fresh empty list	Default list is shared between calls	Use `basket=None`, then create the list inside	High	*fill in*
E6	any	Outdated syntax	`print "hello"` (no brackets)	SyntaxError in Python 3	`print("hello")`	Medium	not observed
E7	any	Outdated syntax	`raw_input()`, `xrange()`	NameError in Python 3	Use `input()`, `range()`	Medium	not observed
E8	any	Unsafe code	`x = eval(input(...))`	Runs anything the user types as code	`int(input(...))` or `float(...)`	High	not observed
E9	any	Unsafe code	`os.system('rm -rf ' + input())`	Lets input run shell commands	Avoid; use file functions with validated paths	High	not observed
E10	any	Poor practice	`f = open(...)` without closing	File may stay open	Use `with open(...) as f:`	Low	not observed

Summary

- Confirmed wrong claims: 5 (E1–E5), all caught by running code.
- Pattern: mistakes cluster around **boundaries** (range, slices), **modes** (w vs a) and **hidden behaviour** (default arguments).
- Risky patterns the scanner detects: 5 (E6–E10).

- Lesson: an AI explanation can sound confident and still be wrong; running the code is the only reliable check.